

Vinay Santosh S

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Objective

To become a vaild resource that the company has invested on and to bring something new to the table that helps the company to reach its vission. Iam a fresher and I can be moulded into ways the company needs me to be in.

Education

Course / Degree	School / University	Grade / Score	Year
B.E Mechanical	St.Josephs's College of Engineering, OMR	8.15 cgpa	2024
12th	Atomic Energy Central School Anupuram	80%	2020
10th	Atomic Energy Central School Anupuram	76.4%	2018

Skills

- Designing using SolidWorks
- AutoCAD Modeling
- CATIA v5 designing
- Planning using Microsoft Excel
- Microsoft Word
- Good Communicative, listening and writing skills
- Software aided finite element analysis using Abaqus
- Software aided finite element analysis using Ansys
- Hindi speaking skills
- Hindi reading skills
- English speaking, reading and writing skills
- Tamil speaking, reading and writing skills
- Clickup software
- Theories of Constraints

Projects

• SAE - Drone development

A team of 10 members that competed in an SAE event. The task task was to come up with a design and fabricated working model of a drone.

The task can be separated into design, fabrication and flight testing. We primarily used SolidWorks for the modelling. I was involved in the wing design of the drone. Separate parts were designed and assembled later. The analysis was performed for finding whether the structure is intact.

Our drone was presented and went through various rounds such as presentation, inspection- to see whether it satisfies the said parameters, Assembly time and finally flight test

We were able to secure 8th place on the southern region even though we came short.

• Mini project- Design and fabrication of pneumatic sheet metal cutter

Me and my partner designed and fabricated a sheet metal cutter that uses pneumatic cylinder to generate pneumatic power to cut sheet metal. It consists of two blades connected as a linear joint. One blade was connected with the pneumatic cylinder while the other was stationary. When the cylinder was activated . The blades came in contact and cut the metal sheet.

• Research on enhancing road car aerodynamics through parametric design optimization of aerodynamic attachments

This study introduces a comprehensive approach to optimize road car aerodynamics using ANSYS Workbench

coupled with response surface optimization techniques, specifically genetic aggregation. With a primary focus on maximizing key performance metrics including drag reduction, stability, enhancement, overall vehicle efficiency.

Achievements & Awards

- Certified in CATIA V5
- Wrote JEE exams in Architecture (70 percentile) and in Engineering (68 percentile)
- SAE certificate on Drone development
- Certified Grade A2 in Business English by Cambridge (BEC)
- Completed 12th with 80% (CBSE board)
- Completed 10th with 75% (CBSE)

Publications

- **Enhancing Road car aerodynamics through parametric design optimization of aerodynamic attachments**
This study introduces a comprehensive approach to optimize road car aerodynamics using ANSYS Workbench coupled with response surface optimization techniques, specifically genetic aggregation. With a primary focus on maximizing key performance metrics including drag reduction, stability enhancement, and overall vehicle efficiency, the investigation centers around three crucial parameters: the AoA of the wing, wing gap, and number of VGs. Beginning with meticulous definition of objectives and constraints, a detailed baseline model of the vehicle geometry is developed within the ANSYS Workbench environment to capture its aerodynamic intricacies. By harnessing the capabilities of ANSYS Workbench and genetic aggregation for response surface optimization, this study provides an efficient framework for significant advancements in road car aerodynamics, potentially enhancing vehicle performance, efficiency, and overall driving experience.

Certifications

- 1. Certified in CATIA V5
- 2. Certified in Business English by BEC
- 3. Certified in 4D printing and Automation
- 4. Certified in Effective Data presentation with Gen AI Tools
- 5. Lean Six Sigma green belt

Internships

- **FEM using Abaqus in IGCAR Kalpakkam**
I interned for IGCAR Kalpakkam for 10 days. I was tasked with analysing of various components using Abaqus. They were simple components but I learned the foundation of Abaqus